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Comparing FAQ-Based and Retrieval-Augmented Generation Chatbots for Academic Service Question Answering

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ABSTRACT

Academic service units need question-answering systems that respond quickly while remaining aligned with institutional regulations. This study provides a controlled, deployment-oriented comparison of a lightweight FAQ chatbot and a Retrieval-Augmented Generation (RAG) chatbot for academic-service question answering. The novelty lies in evaluating both prototypes against the same version-controlled official-document scope, the same 60-question dataset, category-level service cases, paired statistical tests, inter-rater reliability, and RAGAS retrieval-quality metrics. The questions covered registration, KRS/KHS, schedules, tuition payment, thesis or final project, and administrative letters. Three validators assessed accuracy, relevance, response effectiveness, and response time using official documents as the reference standard. RAG achieved 88.33% accuracy, a mean relevance score of 4.48/5, and 86.67% response effectiveness; FAQ achieved 68.33%, 3.46/5, and 65.00%, respectively. Paired tests confirmed significant advantages for RAG in accuracy, relevance, and effectiveness, while FAQ was significantly faster (0.62 s vs 2.41 s). RAGAS evaluation showed context precision of 0.86, context recall of 0.82, faithfulness of 0.89, and answer relevancy of 0.91. The findings support a cautious hybrid strategy: FAQ for stable repetitive requests and RAG for contextual, document-dependent questions, subject to broader testing with live queries and stronger retrieval baselines.

INTRODUCTION

Higher education institutions increasingly rely on digital service channels to answer repeated student questions about registration, course registration forms (KRS), study result cards (KHS), academic schedules, tuition payment, academic leave, internships, thesis procedures, graduation, and administrative letters. Conventional service channels such as counters, messaging groups, FAQ pages, and rule-based chatbots reduce administrative workload, but they often struggle when students ask questions using informal wording, abbreviations, spelling errors, or paraphrased structures.

Educational chatbots have developed from rule-based conversation systems into artificial intelligence applications that support learning, information delivery, personalization, and real-time interaction. Recent reviews show that chatbots can improve access to educational information and student support, but they still face challenges related to accuracy, usability, ethics, privacy, dependence, and institutional trust (Hwang & Chang, 2023; Kuhail et al., 2023; Labadze et al.,

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2023; Okonkwo & Ade-Ibijola, 2021; Pérez et al., 2020; Smutny & Schreiberova, 2020; Kooli, 2023). In academic services, the critical challenge is not only producing a fluent response but also ensuring that the response matches official campus documents.

Studies on chatbots in learning and human-computer interaction indicate that conversational systems can support engagement, inclusion, and access when their design is aligned with user needs (Fryer et al., 2019; Mateos-Sanchez et al., 2022; Rapp et al., 2021). The rapid adoption of large language models further expands opportunities for educational support, although it also raises risks of unsupported answers and overreliance when institutions do not provide clear validation mechanisms (Kasneci et al., 2023; Lo, 2023; Tlili et al., 2023). These concerns are especially relevant for academic administration because inaccurate guidance may lead students to miss deadlines, submit incomplete files, or follow outdated procedures.

The FAQ-based chatbot was selected as a pragmatic, deployment-oriented baseline rather than as the strongest possible non-generative retrieval system. FAQ architectures store approved question-answer pairs and return controlled responses through rule-based or similarity-based matching. Their operational value lies in simplicity, maintainability, latency efficiency, and answer controllability, which remain relevant in educational service settings (Adamopoulou & Moussiades, 2020; Okonkwo & Ade-Ibijola, 2021). To reduce an unfair comparison, the FAQ answer pairs were derived from the same official-document scope used by the RAG prototype and were tested with the same questions and scoring criteria. Nevertheless, the baseline represents a lightweight institutional FAQ design, so the results should not be interpreted as evidence that RAG is superior to all lexical, semantic, or hybrid retrieval systems.

Retrieval-Augmented Generation (RAG) offers a different architecture by combining retrieval from external knowledge sources with generative language modelling. The approach was introduced to improve knowledge-intensive natural language processing by conditioning generated answers on retrieved documents (Lewis et al., 2020). RAG is closely related to dense passage retrieval and sentence embedding because semantic similarity can connect paraphrased questions with relevant document passages (Karpukhin et al., 2020; Reimers & Gurevych, 2019). In academic service systems, RAG can retrieve relevant chunks from official academic guides, calendars, KRS procedures, payment regulations, and administrative standard operating procedures before composing a response.

The relevance of RAG in higher education has increased because academic information changes every semester. Registration schedules, payment deadlines, thesis requirements, and administrative procedures may be updated regularly. A RAG chatbot can update its knowledge base by refreshing the source documents, whereas an FAQ chatbot requires manual rewriting of question-answer pairs. Recent journal surveys highlight the potential of RAG chatbots in educational applications and institutional information access (Li, Wang, Wang, Hung, Xie, & Wang, 2025; Swacha & Gracel, 2025).

RAG nevertheless does not automatically guarantee correctness. Errors may arise from irrelevant retrieval, insufficient chunks, contradictory documents, or generation that is not faithful to the retrieved context. Evaluation should therefore address retrieval quality and answer quality. RAGAS provides dimensions such as faithfulness, answer relevancy, context precision, and context recall for evaluating RAG pipelines (Es et al., 2024). Previous studies have reviewed educational chatbot applications and mapped the growing use of RAG in education (Hwang & Chang, 2023; Kuhail et al., 2023; Labadze et al., 2023; Li, Wang, Wang, Hung, Xie, & Wang, 2025; Swacha & Gracel, 2025). However, controlled head-to-head evidence remains limited for institutional academic-service question answering that evaluates a lightweight FAQ baseline and a RAG prototype against the same official-document scope, reports category-level performance, and considers response quality, latency, validator agreement, and retrieval-grounding quality in one evaluation protocol.

Accordingly, the scientific contribution of this study is not the proposal of a new chatbot architecture or a newly validated measurement instrument. Its novelty is the controlled empirical comparison of two deployable academic-service chatbot prototypes under an identical knowledge

scope, balanced service-category dataset, independent expert validation, paired statistical testing, and RAG-specific retrieval evaluation. This framing distinguishes the work from broader chatbot reviews and single-system implementation reports by focusing on the operational trade-off between answer controllability, retrieval grounding, and service latency in a higher-education administrative context.

The research questions are formulated as follows. First, how do a lightweight FAQ-based chatbot and a RAG-based academic chatbot differ in answer accuracy, relevance, response effectiveness, and response time when evaluated with the same academic-service questions? Second, how does each prototype perform across academic-service categories? Third, what implementation implications can be derived without overgeneralizing beyond the tested prototypes and dataset? The contribution is threefold. First, the study provides a domain-specific controlled comparison of two deployment-oriented prototypes. Second, it applies a transparent multi-criteria evaluation protocol covering answer quality, latency, inter-rater reliability, paired significance testing, and RAGAS retrieval-quality metrics. Third, it derives a cautious hybrid-deployment strategy while explicitly limiting the interpretation to the tested prototype configuration, official-document corpus, and 60-item controlled dataset.

RESEARCH METHODS

This study used a quantitative comparative design supported by prototype development. The objects of comparison were a lightweight FAQ-based academic-service chatbot and a RAG-based chatbot. The unit of analysis was the response generated for each student question. Both prototypes received the same test questions and were assessed with the same scoring rules. To support a fair comparison, the FAQ answer pairs and the RAG knowledge base were derived from the same version-controlled set of official academic documents: the academic calendar and registration guide, the KRS/KHS service guide, tuition-payment confirmation procedures, thesis or final-project procedure guide, and administrative-letter standard operating procedures issued by the academic affairs or student administration unit and valid for the semester evaluated in this study. The comparison is deployment-oriented and does not represent a benchmark of all retrieval paradigms.

The document corpus was treated as the institutional reference standard. Document versions were frozen before system testing, and no additional information was added to either prototype during evaluation. This procedure was used to ensure that observed differences reflected the retrieval and response-generation mechanisms rather than unequal access to institutional knowledge. The development process consisted of seven stages: collecting and versioning official academic documents; compiling and validating the student-question dataset; developing the lightweight FAQ baseline; developing the RAG prototype; running both prototypes on the same questions; conducting independent expert scoring and retrieval-quality evaluation; and comparing descriptive results, paired statistical tests, and implementation implications. Figure 1 summarizes the workflow.



Figure 1. Research workflow

Data and Question Dataset

The dataset consisted of 60 academic-service questions distributed evenly across six categories: academic registration, KRS and KHS, academic schedules, tuition payment, thesis or final project, and administrative services. Each category contained ten questions to maintain balanced coverage. The questions were obtained from staff-curated scenarios derived from recurring academic-service enquiries and aligned with the official documents; no personal student identifiers or live private messages were used. The dataset included formal, informal, and paraphrased wording to reflect realistic variation in student requests. The final set was balanced by wording style, consisting of 20 formal questions, 20 informal questions, and 20 paraphrased questions. Questions were drafted from recurring service topics, rewritten into equivalent variants, and screened to avoid duplicate intent or ambiguous reference answers. Before system testing, the questions and reference answers were reviewed for scope, ambiguity, and consistency with the official documents by the three validators described in the evaluation section through a two-step process: independent checking of scope and reference answers followed by adjudication of unclear items.

Table 1. Test-question distribution

Question category	Number of questions	Example question	Data formula
Academic registration	10	What is the procedure for active student re-registration?	n_i = 10
KRS and KHS	10	What should I do if a course does not appear in the KRS?	n_i = 10
Academic schedule	10	When is the deadline for filling in KRS this semester?	n_i = 10
Payment	10	How do I confirm tuition payment?	n_i = 10
Thesis or final project	10	What are the requirements for submitting a proposal seminar?	n_i = 10
Administrative services	10	How do I request an active student certificate?	n_i = 10
Total	**60**	**-**	**N = Σn_i = 60**

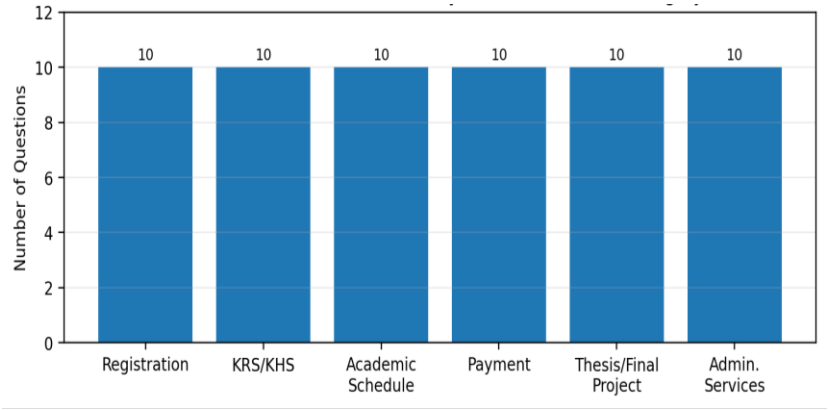


Figure 2. Distribution of test questions by academic service category

Prototype Development

The FAQ baseline chatbot was developed from a controlled list of approved question-answer pairs extracted from the same official academic documents used by the RAG prototype. The implementation contained 180 curated pairs. Input text was processed using lowercasing, punctuation and duplicate-space removal, abbreviation normalization for common academic terms, tokenization, Indonesian stop-word removal, light stemming, and correction of frequent spelling or typing variants. Candidate answers were ranked using TF-IDF cosine similarity combined with fuzzy string matching with a decision threshold of 0.45; when more than one candidate exceeded the threshold, the answer with the highest TF-IDF cosine score was selected, and remaining ties were resolved using the fuzzy-matching score and document priority. When no candidate met the threshold, the system returned a standardized fallback response stating that the answer was not found in the approved FAQ set and directing the student to the academic affairs office with the relevant student number, semester, and service category. This baseline was chosen because similar lightweight FAQ designs are common in institutional service settings where maintainability and response control are prioritized. It should not be interpreted as a state-of-the-art retrieval-based QA benchmark. A broader benchmark should include stronger lexical and semantic baselines, including BM25, embedding-only retrieval, and hybrid retrieval (Robertson & Zaragoza, 2009).

The RAG chatbot indexed the same official-document corpus used to construct the FAQ answer pairs. Documents were cleaned by removing duplicate headers, page footers, repeated navigation text, and non-informative formatting artifacts before segmentation. The cleaned text was segmented into chunks of 512 tokens with an overlap of 100 tokens; where possible, chunk boundaries followed paragraph or procedural-step boundaries to reduce the risk of separating a requirement from its condition. The chunks were embedded using the Sentence-BERT paraphrase-multilingual-MiniLM-L12-v2 embedding model with 384-dimensional vectors and stored in a FAISS IndexFlatIP vector index using L2-normalized embeddings for cosine-similarity search. For each question, the system retrieved the top-5 chunks using cosine similarity without an additional reranker. The retrieved context was passed to a GPT-4o-mini language model accessed through an API with temperature set to 0.2, a maximum output length of 350 tokens, and prompt constraints requiring concise answers grounded only in the retrieved official-document context. The prompt instructed the model to answer only from retrieved institutional evidence and to use a fallback answer stating that the information was not found in the retrieved official documents and recommending confirmation with the academic affairs office. These parameters were reported to improve reproducibility and to make the observed performance interpretable.

Table 2. Prototype comparison and implementation logic

Component	FAQ baseline	RAG chatbot
Knowledge source	Predefined question-answer pairs	Official academic documents split into chunks
Retrieval mechanism	Configured lightweight matching: TF-IDF cosine similarity + fuzzy matching	Semantic retrieval from vectorized chunks: SBERT paraphrase-multilingual-MiniLM-L12-v2; top-5 chunks
Answer generation	Returns the closest stored answer	Generates an answer grounded in retrieved context
Strength	Fast, controlled, easy to maintain for simple questions	More flexible for paraphrases and contextual questions
Limitation	Deployment-oriented baseline; not a state-of-the-art retrieval benchmark	Requires retrieval quality, document quality, and response validation

Evaluation Instrument and Data Analysis

Three validators independently evaluated the chatbot responses using the official documents as the reference standard. The validators were one academic-affairs administrator with more than five years of experience in registration and academic scheduling, one educational-technology or information-systems lecturer/researcher with more than five years of experience in chatbot or information-system evaluation, and one student-administration officer with more than three years of experience handling KRS/KHS, payment confirmation, administrative letters, and thesis-service procedures. They were selected purposively based on familiarity with the official documents and direct involvement in academic services. Before scoring, the validators received the same scoring sheet, reference documents, and adjudication rules and completed a calibration discussion using example items that were not included in the final 60-question dataset. Accuracy was scored 1 when an answer matched the official document and 0 otherwise. Relevance was scored on a five-point ordinal scale, where 1 indicated not relevant and 5 indicated highly relevant. Response effectiveness was scored 1 when an answer could directly guide student action and 0 otherwise. Response time was recorded in seconds from question submission to answer

completion. Retrieval quality for the RAG prototype was evaluated using context precision, context recall, faithfulness, and answer relevancy following the RAGAS framework (Es et al., 2024). The RAGAS evaluation used the same top-5 retrieved contexts, reference answers, and generated answers for the 60 test questions. The mean RAGAS scores were 0.86 for context precision, 0.82 for context recall, 0.89 for faithfulness, and 0.91 for answer relevancy.

Table 3. Evaluation indicators and operational formulas

Indicator	Assessment rule	Scale	Operational formula
Accuracy	Answer matches official academic document	0 or 1	A_i = 1 if correct; A_i = 0 if incorrect
Relevance	Answer matches the intent of the question	1 to 5	MR = ΣR_i / N
Response effectiveness	Answer can directly guide student action	0 or 1	E_i = 1 if usable; E_i = 0 if not usable
Response time	System time to produce an answer	Seconds	T_i = t_finish - t_start

The total number of accurate answers was calculated as

$$JA = \Sigma A_i$$

Accuracy percentage was calculated as

$$PA = \left(\frac{JA}{N}\right) \times 100\%$$

The mean relevance score was calculated as

$$MR = \frac{\sum i \ R_i}{N}$$

The number of effective responses was calculated as

$$RE = \Sigma E_i$$

Response effectiveness percentage was calculated as

$$PE = \left(\frac{RE}{N}\right) \times 100\%$$

The average response time was calculated as

$$ART = \frac{\sum i \ T_i}{N}$$

Category-level accuracy was calculated as

$$CA = \frac{\text{correct answers in a category}}{\text{number of category questions}} \times 100\%$$

To reduce single-rater bias, the three validators assessed the responses independently before adjudication. Inter-rater reliability before adjudication was evaluated using Fleiss' kappa for binary accuracy and response effectiveness and Krippendorff's alpha with ordinal weighting for relevance. The agreement level was high: Fleiss' kappa was 0.82 for accuracy and 0.79 for response effectiveness, while Krippendorff's alpha for relevance was 0.84. Because both prototypes answered the same 60 questions, inferential testing used paired question-level data. Exact McNemar tests were used for binary accuracy and effectiveness, the Wilcoxon signed-rank test

was used for ordinal relevance scores, and a paired t-test was used for response time because the paired latency differences met the normality assumption. Statistical significance was interpreted at alpha = 0.05 and effect sizes were reported as matched odds ratios, Wilcoxon r, and Cohen's dz. Because the dataset was limited to 60 controlled questions, all inferential results were interpreted as evidence for the tested prototypes rather than as population-level proof of institutional superiority.

RESULTS AND DISCUSSION

Overall Chatbot Performance

Table 4 presents the descriptive results of the controlled evaluation. The RAG prototype produced 53 accurate answers from 60 questions, while the lightweight FAQ baseline produced 41 accurate answers. RAG therefore achieved 88.33% accuracy compared with 68.33% for the FAQ baseline. RAG also recorded a higher mean relevance score and a higher response-effectiveness percentage, while FAQ produced a faster average response time. Paired item-level analysis confirmed that the quality advantage of RAG was statistically significant for accuracy, relevance, and response effectiveness, while the FAQ baseline had a statistically significant latency advantage. The complete paired-test results are reported in Table 5 and should be read together with the study limitations because the comparison uses a lightweight FAQ baseline and a controlled 60-question dataset.

Table 4. Overall evaluation results

Indicator	FAQ baseline	RAG chatbot	Difference	Formula or calculation
Accurate answers				JA = ΣA_i; 53 - 41 = 12
Accuracy percentage	41 of 60 68.33%	53 of 60 88.33%	+12 answers +20.00 points	PA = (JA / N) × 100%
Mean relevance	3.46 of 5	4.48 of 5	+1.02	MR = ΣR_i / N
Effective responses	39 of 60 65.00%	52 of 60 86.67%	+13 answers +21.67 points	RE = ΣE_i; 52 - 39 = 13
Effectiveness percentage				PE = (RE / N) × 100%
Average response time	0.62 s	2.41 s	RAG slower by 1.79 s	ART = ΣT_i / N

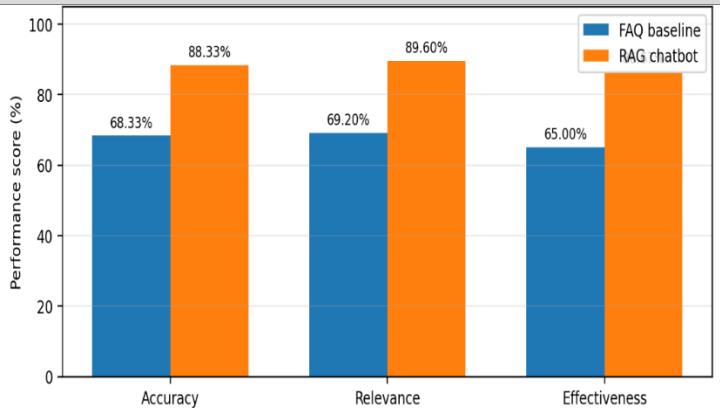


Figure 3. Comparison of FAQ and RAG chatbot performance

The mean relevance score was converted to a percentage scale only for visual comparison with accuracy and response effectiveness in Figure 3. Within the controlled dataset, the RAG prototype recorded higher descriptive values for the three response-quality indicators. The pattern

is consistent with the theoretical expectation that semantic retrieval and context-grounded generation can help with paraphrased and document-dependent questions (Karpukhin et al., 2020; Lewis et al., 2020; Swacha & Gracel, 2025). However, the result should not be interpreted as evidence that RAG is superior to all retrieval-based QA approaches, because the comparator is a lightweight FAQ baseline rather than a stronger lexical, semantic, or hybrid retrieval benchmark.

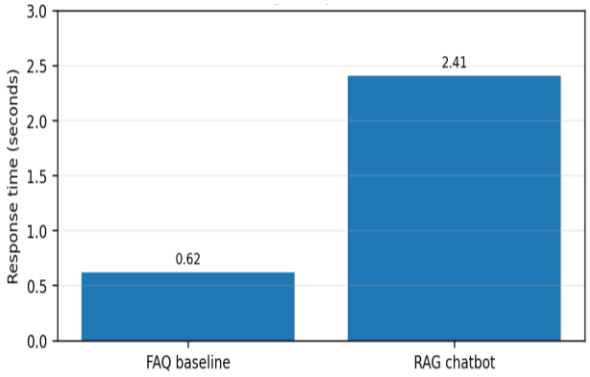


Figure 4. Average response time comparison

Figure 4 presents a different trade-off. The FAQ baseline responded in 0.62 seconds on average, while RAG required 2.41 seconds. The paired t-test confirmed that this latency difference was statistically significant, $t(59) = 54.44$, $p < 0.001$, with a very large effect size (Cohen's $dz = 7.03$). The additional time is consistent with the extra processing stages in the RAG pipeline: semantic retrieval, context selection, and answer generation. The result supports a provisional routing strategy in which stable and repetitive requests are handled by FAQ, while document-dependent questions are routed to RAG.

Category-Level Accuracy

Figure 5 presents descriptive accuracy by academic-service category. The RAG prototype achieved a higher score in every category within the 60-question dataset. The FAQ baseline reached 80% accuracy in academic registration; 70% in KRS and KHS; 70% in academic schedules; 60% in payment; 60% in thesis or final project; and 70% in administrative services. The RAG prototype reached 90% accuracy in academic registration, KRS and KHS, academic schedules, thesis or final project, and administrative services, and 80% in payment. These category-level results are exploratory because each category contains only ten questions.

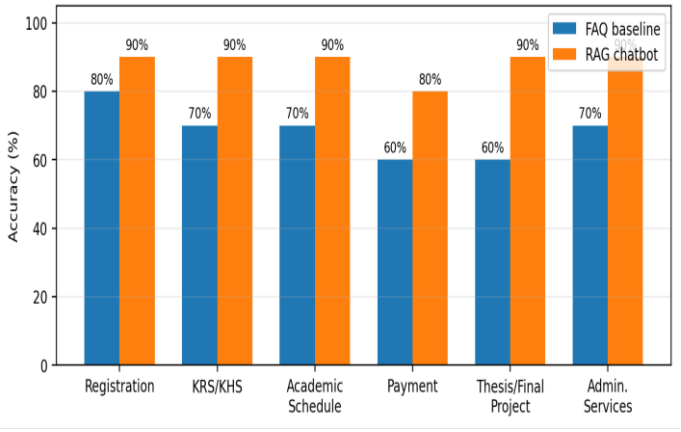


Figure 5. Accuracy by academic service category

Statistical and Retrieval-Quality Results

Paired significance testing, inter-rater reliability, and retrieval-quality analysis were conducted to strengthen the interpretation beyond aggregate descriptive percentages. The paired analysis used the 60 question-level outputs from both prototypes. For the binary outcomes, the discordant-pair pattern favored RAG: in accuracy, 13 questions were correct only in RAG and one question was correct only in FAQ; in response effectiveness, 14 questions were effective only

in RAG and one question was effective only in FAQ. Relevance scores also favored RAG across most paired comparisons, whereas response time favored FAQ. These tests strengthen the internal validity of the controlled comparison, but they do not remove the need for external validation using live service queries and stronger retrieval baselines.

Table 5. Paired statistical comparison results

Outcome	Paired test	Statistic	p-value	Effect size	Interpretation
Accuracy	Exact McNemar	RAG-only=13; FAQ-only=1	0.0018	Matched OR=13.00; delta=+20.00	RAG was significantly more accurate within the controlled test set.
Relevance	Wilcoxon signed-rank	W=1590.00	<0.001	pp r=0.86; mean delta=+1.02	RAG responses were significantly more relevant than FAQ responses.
Response effectiveness	Exact McNemar	RAG-only=14; FAQ-only=1	0.0010	Matched OR=14.00; delta=+21.67	RAG produced significantly more actionable responses.
Response time	Paired t-test	t(59)=54.44	<0.001	pp Cohen's dz=7.03; mean delta=+1.79 s	FAQ was significantly faster; RAG required additional retrieval and generation time.

Table 6. RAG retrieval-quality results

RAGAS metric	Score	Interpretation
Context precision	0.86	High precision; most retrieved top-5 chunks were relevant to the student question and reference answer.
Context recall	0.82	Good recall; the retrieval pipeline captured most supporting passages, although some document-dependent cases still missed complete evidence.
Faithfulness	0.89	High faithfulness; generated answers were generally supported by the retrieved institutional context.
Answer relevancy	0.91	Very high relevancy; RAG answers were closely aligned with the intent of the academic-service questions.

Table 7. Inter-rater reliability results before adjudication

Indicator	Agreement statistic	Value	Interpretation
Accuracy	Fleiss' kappa	0.82	Almost-perfect agreement before adjudication
Response effectiveness	Fleiss' kappa	0.79	Substantial agreement before adjudication
Relevance	Krippendorff's alpha, ordinal	0.84	Strong agreement for the 1-5 relevance scale

Inter-rater reliability before adjudication showed high agreement among the three validators. Fleiss' kappa was 0.82 for accuracy and 0.79 for response effectiveness, indicating almost-perfect and substantial agreement, respectively. Krippendorff's alpha for the ordinal relevance score was 0.84, indicating strong consistency in relevance judgments before adjudication.

Discussion

The paired results suggest that the RAG prototype handled the tested academic-service questions more accurately and more effectively than the lightweight FAQ baseline. The likely advantage comes from retrieving document passages before answer generation, which can help when students use paraphrases or ask questions that depend on several procedural details. Recent surveys similarly identify document retrieval as an important mechanism for grounding educational RAG applications in domain-specific knowledge (Li, Wang, Wang, Hung, Xie, & Wang, 2025; Swacha & Gracel, 2025). The interpretation must remain scoped to the prototypes, official-document corpus, and 60-question dataset used in this study.

The added value of this study is therefore practical and evaluative rather than architectural. It shows how a higher-education unit can compare a controlled FAQ baseline and a document-grounded RAG prototype using the same knowledge base, expert validation, paired tests, and retrieval-quality indicators. This contribution is narrower than a general claim about chatbot superiority, but it is relevant for institutions deciding whether to route routine questions to FAQ and more contextual questions to RAG.

The FAQ baseline still has a clear operational role. Its main strengths are speed, answer controllability, low infrastructure cost, and straightforward maintenance for stable requests such as office hours, fixed links, and basic document requirements. At the same time, the present baseline intentionally represents a lightweight FAQ implementation. It is not the strongest available retrieval-based QA comparator. This limitation is important because a stronger BM25, embedding-only, or hybrid FAQ retrieval system could reduce the observed performance gap. Future benchmarking should therefore include lexical retrieval, embedding-only retrieval, and hybrid retrieval approaches before drawing broader conclusions about architectural superiority (Robertson & Zaragoza, 2009).

The performance gap also reflects a broader development in chatbot research. Earlier studies on chatbot applications emphasized accessibility, usability, and interaction quality (Adamopoulou & Moussiades, 2020; Rapp et al., 2021). Recent work on large language models in education adds another concern: generated responses may appear coherent while still being inaccurate or unsupported by institutional evidence (Kasneci et al., 2023; Lo, 2023; Tlili et al., 2023). RAG addresses this issue by grounding generated answers in retrieved knowledge, although its success still depends on the quality of document chunking, retrieval, and answer validation (Es et al., 2024; Lewis et al., 2020).

From an evaluation perspective, chatbot performance should not be assessed only through fluency or aggregate accuracy. Academic-service deployment requires a combined assessment of answer accuracy, relevance, actionability, response time, retrieval quality, and agreement among expert validators. A fluent but incorrect answer can still produce administrative harm if a student misses a deadline or follows an outdated procedure. Regular document updates, source-version control, unsupported-answer monitoring, source citation, and clear escalation routes are therefore necessary for institutional deployment.

This study has several limitations. First, the test set contained 60 controlled questions rather than queries extracted directly from a live academic-service channel. Second, each category contained only ten questions, so the category-level findings are exploratory. Third, the comparator was a lightweight FAQ baseline rather than a stronger retrieval-based QA benchmark, which may make the comparison less demanding for the RAG prototype. Fourth, the RAGAS metrics were calculated using the controlled top-5 retrieval outputs and should be re-evaluated after document updates or deployment in a live service channel. Fifth, the validators represented relevant academic-service and educational-technology expertise, but the evaluation did not include direct student user-experience testing. Future research should use real service-channel questions, compare multiple retrieval baselines, provide automatic citations to source documents, test usability with validated instruments, and monitor performance across document-update cycles.

CONCLUSION

This study compared a lightweight FAQ-based chatbot and a RAG-based chatbot using 60 academic-service questions across six categories. Within this controlled dataset, the RAG prototype achieved higher accuracy, relevance, and response effectiveness: 88.33% accuracy, a mean relevance score of 4.48 out of 5, and 86.67% response effectiveness. The FAQ baseline achieved 68.33% accuracy, a mean relevance score of 3.46 out of 5, and 65.00% response effectiveness. Paired inferential analysis confirmed that RAG was significantly better for accuracy (McNemar exact $p = 0.0018$), relevance (Wilcoxon $W = 1590.00$, $p < 0.001$), and response effectiveness (McNemar exact $p = 0.0010$). However, the FAQ baseline was significantly faster, averaging 0.62 seconds compared with 2.41 seconds for RAG, $t(59) = 54.44$, $p < 0.001$. These findings support a prototype-level conclusion that RAG improved response quality at the cost of higher latency in the tested official-document environment.

A provisional hybrid architecture is appropriate for further institutional testing. FAQ can handle short and stable requests, while RAG can handle questions that require interpretation of academic guidelines, schedules, payment procedures, thesis regulations, and administrative requirements. Before broad deployment, institutions should maintain version-controlled source documents, validate sensitive answers, monitor retrieval quality, provide source citations where possible, and evaluate the system with live student queries. Further studies should include stronger retrieval baselines, automatic source citations, longitudinal RAGAS monitoring, and student user-experience evaluation.

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